

Free-of-Imputation Regression Forest to Handle Missingness in Predicting Individual Treatment Effect

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Predicted Individual Treatment Effects (PITEs)

$$\tau(X_i) = f_1(X_i) - f_0(X_i)$$

- X_i : true covariates
- Y : Outcomes
- $t = 0, 1$: arms
- $f_t(x)$: Predictions under arms

[Lamont et al., 2016]

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Missing covariates

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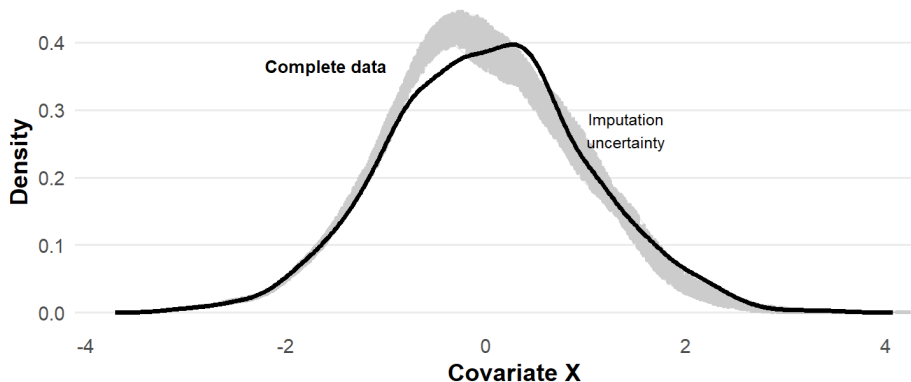
Key point

How can we estimate trustworthy, robust PITE ?

PITE depends on the covariate information available for each patient.

Imputation changes the predictor distribution learned by PITE

Black = complete data | Grey = 30 imputed datasets | band = 10th–90th percentile across imputations



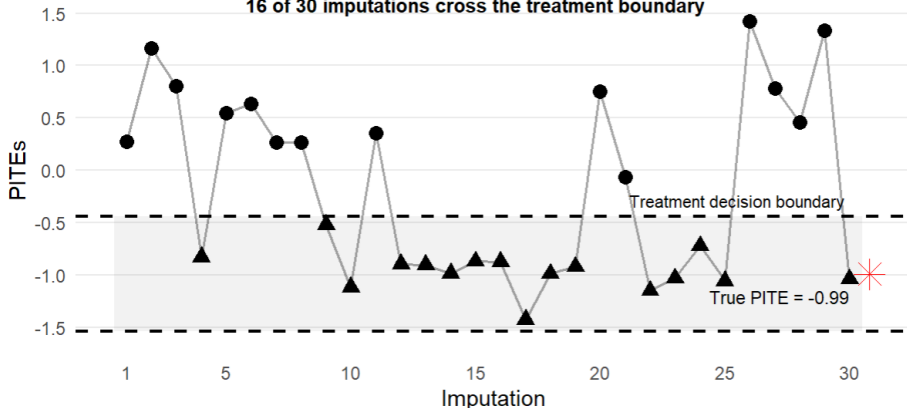
The missing value changes the recommended treatment.

Same patient. Different imputation. Different treatment decision

Patient 15 | Mean PITE = -0.21 | SD = 0.87

Decision ● non Recommended ▲ Recommended

16 of 30 imputations cross the treatment boundary



Imputation Error and PITE

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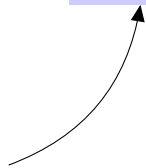
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- The local gradient of the PITE surface is $\nabla \tau(X_i) = \nabla f_1(X_i) - \nabla f_0(X_i)$

Imputation Error: Bias and Variance

Let $\Delta_i = \tilde{X}_i - X_i$.

Bias: systematic imputation error

$$\Gamma_i = E[\nabla\tau(X_i)^\top \Delta_i]$$

- Depends on the **mean and direction** of the imputation error.

Variance: imputation uncertainty

$$B_i = \nabla\tau(X_i)^\top \text{Var}(\Delta_i) \nabla\tau(X_i)$$

- Depends on the **dispersion and covariance** of the imputation error.
- Both effects are governed by the local PITE gradient $\nabla\tau(X_i)$.
- Imputation uncertainty is **amplified by local treatment-effect heterogeneity**.

Innovation: Free-Missingness Approach

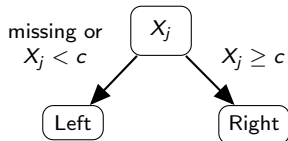
Do not reconstruct the missing value first. Let the learner decide how missingness should affect the prediction.

- Treat missingness as informative features
- Bypass explicit imputation
- Avoid injected noise, bias, or inflated uncertainty

Missingness-Incorporating Splitting in BART

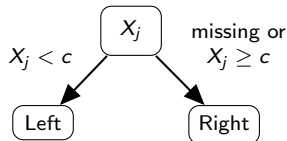
Rule 1

Missing $\rightarrow$ Left



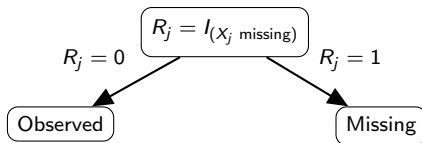
Rule 2

Missing $\rightarrow$ Right



Rule 3

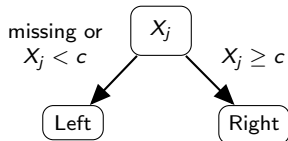
Missingness indicator



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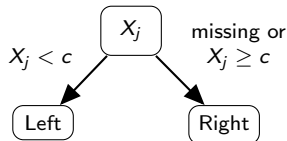
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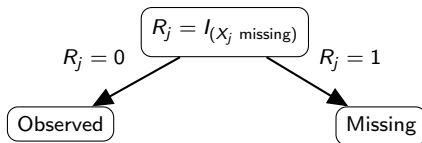
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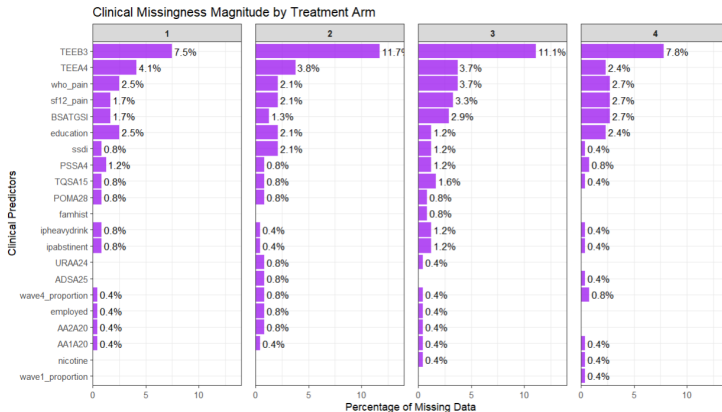
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Error Reduction?

Application



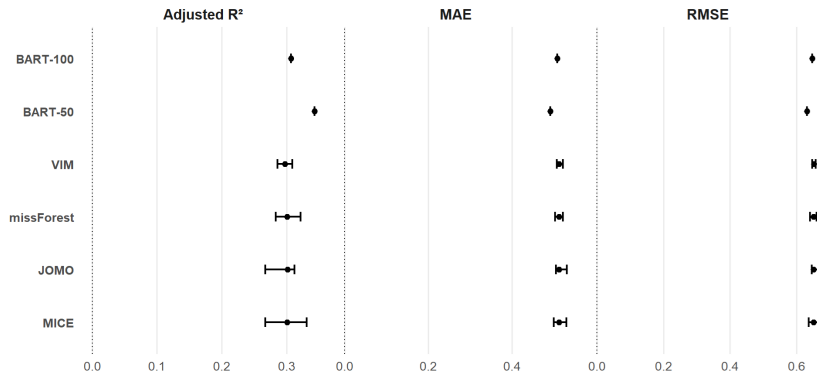
21 of 32 covariates contain missing information (65.6%)

Patients by treatment group: $n_t \approx 240$, $t = 1, \dots, 4$

Results

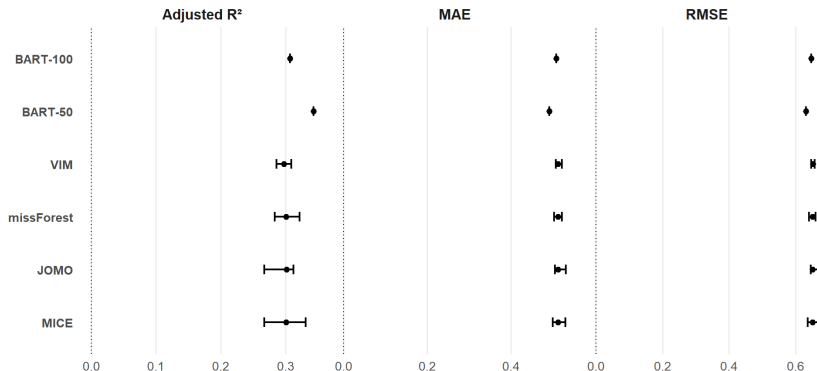
Predictive performance with and without imputation

Points = mean prediction performance; intervals = range across imputations



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Computational burden: for $m = 10$ the imputation time is approximately:

- MICE: 5.60 min
- missForest: 5.51 min
- VIM: 1.27 min
- JOMO: 15.14 hours

Conclusions

- Learning missingness yields lower or equivalent risk than imputation
- Free-missingness is a robust, efficient, and conceptually simple alternative for PITE
- Free-of-imputation BART preserves information
- Tighter uncertainty, computationally efficient
- Robust across heterogeneous clinical settings (MAR)

- Andrea Lamont, Michael D Lyons, Thomas Jaki, Elizabeth Stuart, Daniel J Feaster, Kukatharmini Tharmaratnam, Daniel Oberski, Hemant Ishwaran, Dawn K Wilson, and M Lee Van Horn. Identification of predicted individual treatment effects in randomized clinical trials. *Statistical Methods in Medical Research*, 27(1):142–157, March 2016. ISSN 1477-0334. doi: 10.1177/0962280215623981. URL <http://dx.doi.org/10.1177/0962280215623981>.
- Donald B. Rubin. *Multiple Imputation for Nonresponse in Surveys*. John Wiley & Sons, New York, 1987. doi: 10.1002/9780470316696.
- J. A C Sterne, I. R White, J. B Carlin, M. Spratt, P. Royston, M. G Kenward, A. M Wood, and J. R Carpenter. Multiple imputation for missing data in epidemiological and clinical research: potential and pitfalls. *BMJ*, 338(jun29 1):b2393–b2393, June 2009. ISSN 1468-5833. doi: 10.1136/bmj.b2393. URL <http://dx.doi.org/10.1136/bmj.b2393>.

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